MIT Art Design and Technology University

MIT School of Computing, Pune

Department of Computer Science and Engineering

Lab Manual

Course- Data Modeling & Visualization Laboratory

Class - T.Y. (SEM-V) AIA

Name of the Subject Teacher

Dr. Ranjana Kale

A.Y. 2024 – 2025

Experiment No 6

Experiment Title: Perform Clustering

Problem Statement:

Implement clustering using suitable dataset

Objective: To perform clustering to understand the distribution of the data in dataset

Theory:

Clustering is a data analysis technique that groups similar objects or data points into clusters, where each cluster contains items that are more alike to each other than to items in other clusters. It's a core concept in unsupervised machine learning, meaning it doesn't require predefined labels or categories for the data.

- Unsupervised Learning:

Clustering falls under the umbrella of unsupervised learning because it identifies patterns and structures in data without the need for labeled examples or prior knowledge about the data.

- Similarity:

The process relies on defining a measure of similarity or distance between data points to determine which ones belong to the same cluster.

- Clusters:

Clusters are groups of data points that are more similar to each other than to data points in other clusters.

- Applications:

Clustering is widely used in various fields, including:

  - Customer Segmentation: Grouping customers based on purchasing behavior, demographics, or other characteristics to tailor marketing strategies.
  - Image Analysis: Identifying objects or regions of interest in images.
  - Anomaly Detection: Finding unusual or outlier data points that deviate from the norm.
  - Information Retrieval: Organizing documents or search results into categories.
  - Bioinformatics: Analyzing gene expression data or protein structures.

How it works:

1. Data Input: A dataset with multiple data points or objects is provided as input.
2. Similarity Measure: A suitable measure of similarity or distance is chosen (e.g., Euclidean distance, cosine similarity).
3. Clustering Algorithm: A clustering algorithm is applied to group the data points based on the chosen similarity measure.
4. Cluster Assignment: Each data point is assigned to a cluster.

Common Clustering Algorithms:

- K-Means:

A popular algorithm that partitions data into K clusters, where each point belongs to the cluster with the nearest mean (centroid).

- Hierarchical Clustering:

Builds a hierarchy of clusters, either by merging smaller clusters into larger ones (agglomerative) or by splitting larger clusters into smaller ones (divisive).

- DBSCAN:

A density-based algorithm that groups together data points that are closely packed together, marking as outliers the data points that lie alone in low-density regions.

- Mean Shift:

A centroid-based algorithm that iteratively shifts data points towards the modes (densest regions) of the data distribution.

k-means Clustering

k-Means clustering is an unsupervised machine learning algorithm used for partitioning a dataset into distinct, non-overlapping groups or clusters. The goal is to group similar data points together and separate dissimilar ones. Each cluster is represented by a centroid, which is the average of all data points in that cluster. The algorithm iteratively refines the assignment of data points to clusters and adjusts the centroids to minimize the overall intra-cluster variance.

1. **Key Components:**
   1. **k (Number of Clusters):**
      - The parameter "k" represents the number of clusters the algorithm aims to identify in the dataset. It is a user-defined parameter, and choosing the right value for k is crucial.
   2. **Cluster Centroids:**
      - Each cluster is characterized by a centroid, which is the mean position of all data points in that cluster. Centroids serve as the representatives of their respective clusters.

2. Goal of k-Means Clustering:

The primary goal of k-Means clustering is to minimize the within-cluster sum of squares, often referred to as the **inertia** or **objective function**. The inertia is the sum of the squared distances between each data point and its assigned cluster centroid.

3. Algorithm Workflow:

1. **Initialization:**
   - Select kk initial centroids, either randomly or through a more systematic method.
2. **Assignment:**
   - Assign each data point to the nearest centroid, forming kk clusters.
3. **Update Centroids:**
   - Recalculate the centroids based on the mean position of the data points in each cluster.
4. **Iteration:**
   - Repeat the assignment and centroid update steps until convergence, where the assignments no longer change significantly.
5. **Final Clusters:**
   - The algorithm produces kk clusters, each represented by its centroid.

4. Minimizing Inertia: By assigning data points to clusters and updating centroids iteratively, the algorithm minimizes the inertia, leading to clusters with tight internal cohesion and clear separation between clusters. The final result is a partitioning of the data into groups where points within the same group are more similar to each other than to those in other groups.

5. Steps of the k-Means Algorithm:

- The steps involved in the k-Means algorithm are:
  1. **Initialization:** Select k initial centroids.
  2. **Assignment:** Assign each data point to the nearest centroid.
  3. **Update Centroids:** Recalculate the centroids based on the assigned data points.
  4. **Iteration:** Repeat the assignment and centroid update steps until convergence.

6. Choosing the Number of Clusters (k):

Several methods can be employed to determine the optimal number of clusters, and here are some common approaches: **Elbow Method, Silhouette Score,Gap Statistics and Davies-Bouldin Index**

- **Elbow Method:**
  - **Concept:**
    - The Elbow Method involves running the k-Means clustering algorithm on the dataset for a range of values of k and plotting the sum of squared distances from each point to its assigned center (inertia) against k.
  - **Procedure:**
    - Execute k-Means clustering for a range of k values.
    - Compute the inertia for each k.
    - Plot the inertia values against the corresponding k values.
    - Identify the "elbow" point on the plot where the rate of decrease in inertia slows down.
  - **Interpretation:**
    - The "elbow" is a point where adding more clusters does not significantly reduce the inertia, suggesting diminishing returns in terms of clustering improvement.

7. Distance metrics play a fundamental role in the k-Means clustering algorithm, influencing how data points are assigned to clusters and how centroids are updated during the iterative process. The choice of distance metric affects the shape, size, and characteristics of the clusters formed. Common distance metrics used in k-Means clustering include Euclidean distance, Manhattan distance, and other variations.

8. Handling categorical data in k-Means clustering poses challenges because the algorithm relies on distance metrics, and categorical variables do not have a natural notion of distance.

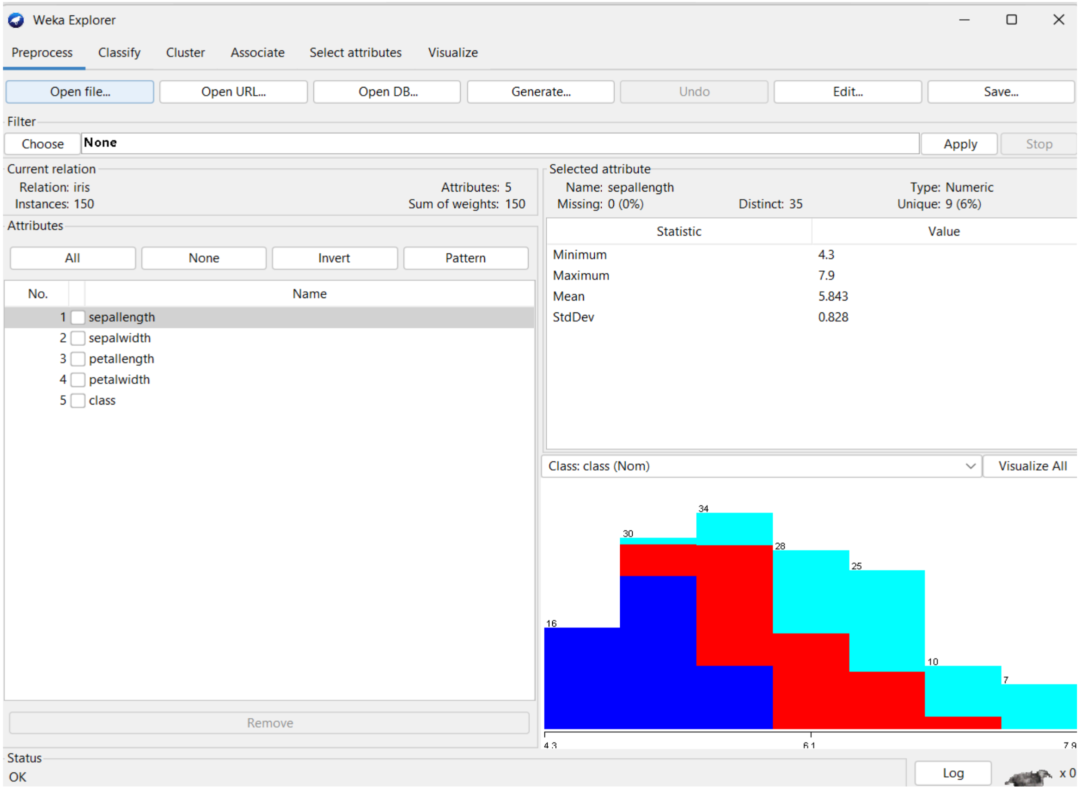
- **Challenges:** No Inherent Distance Measure, Sensitivity to Encoding, Curse of Dimensionality
- **Strategies:** Ordinal Encoding, Target Encoding, Frequency-Based Encoding, Embedding Techniques, K-Prototypes Algorithm, Hybrid Approaches, Dimensionality Reduction, Feature Engineering

Conclusion:

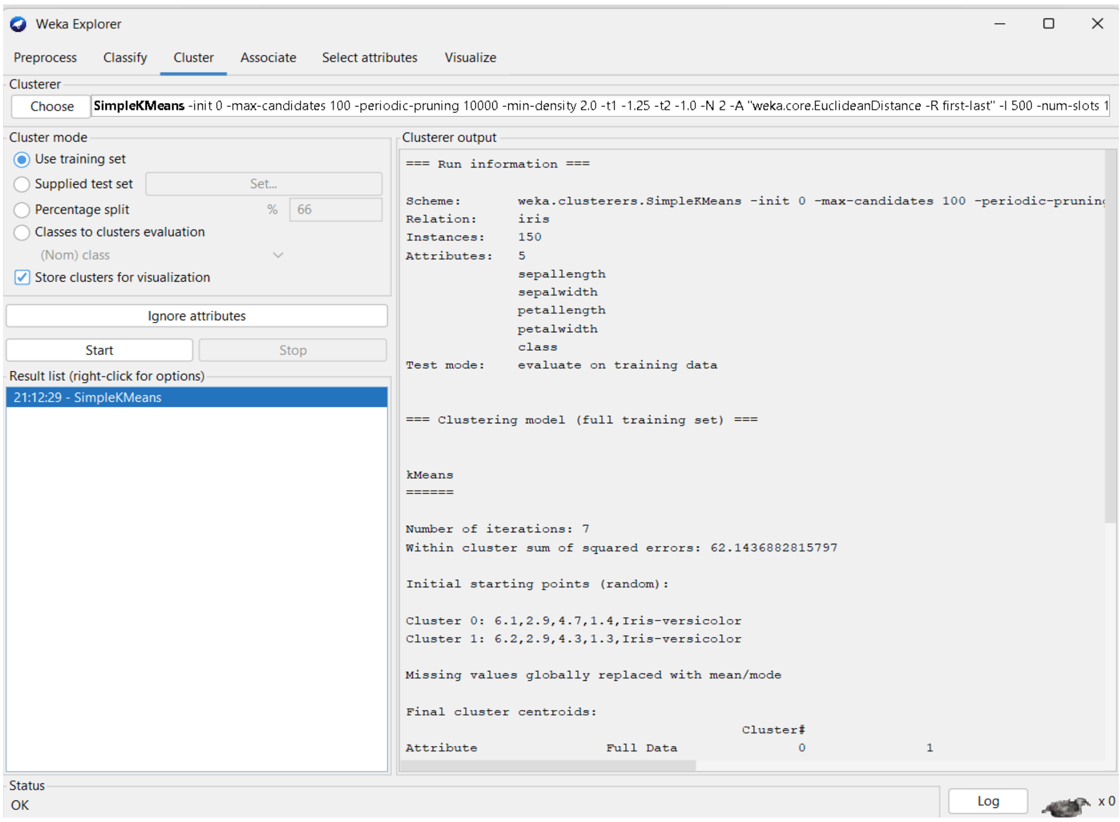
This lab provides students with hands-on experience in implementing and understanding k-means clustering algorithms. It covers fundamental concepts, practical aspects, and considerations for optimal model performance. Students will gain insights into the importance of parameter tuning, distance metrics, and the impact of k on the algorithm's behavior. By focusing on the theoretical aspects, this lab equips students with the foundational knowledge necessary for successful implementation and application of the algorithm in practical scenarios.

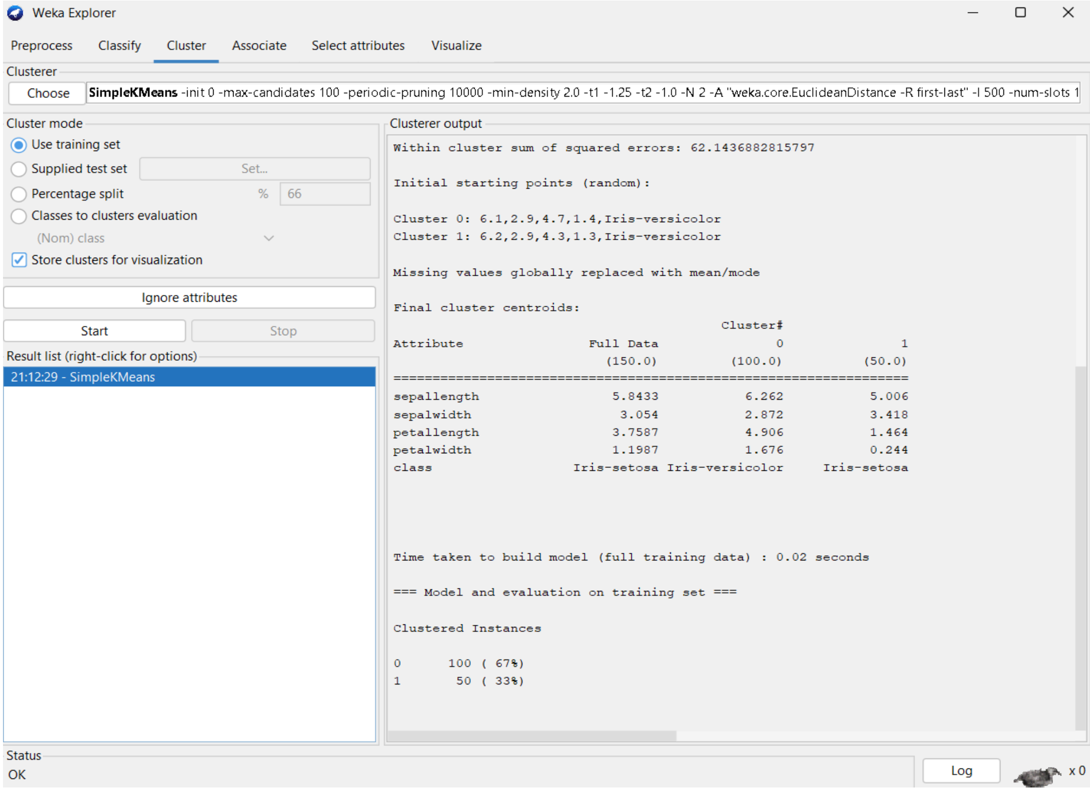
Source Code and Output /Screenshots:

1) Iris dataset opened in Weka

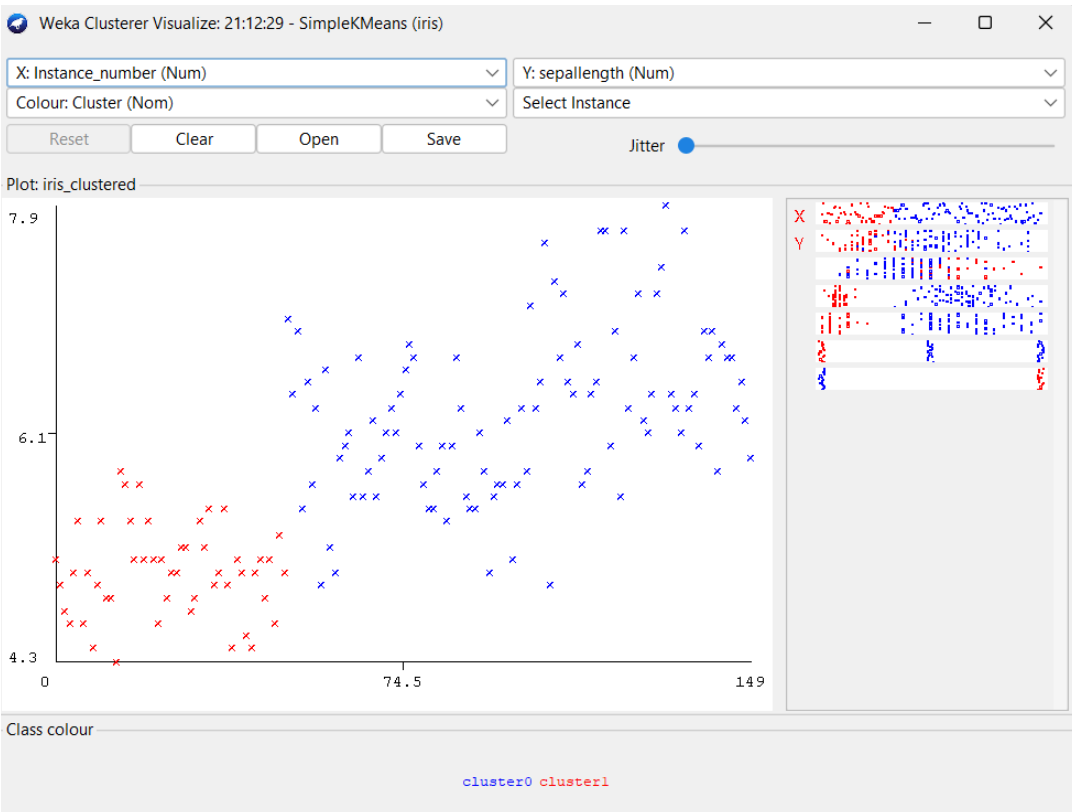


2) Results of Clustering





3) Visualization of Graph



Useful Resource:

https://www.geeksforgeeks.org/machine-learning/k-means-clustering-using-weka/

https://www.tutorialspoint.com/weka/weka_clustering.htm
https://www.youtube.com/watch?v=eVtYfH9LBgk
https://www.youtube.com/watch?v=B8UDaZupCZ8

Exercise Questions

1. What are the different algorithms of clustering in WEKA?
2. Why is selecting the appropriate number of clusters ("k") crucial in k-Means clustering?
3. In what scenarios is feature scaling particularly important in k-Means?
4. How does the centroid update step contribute to the convergence of the algorithm?